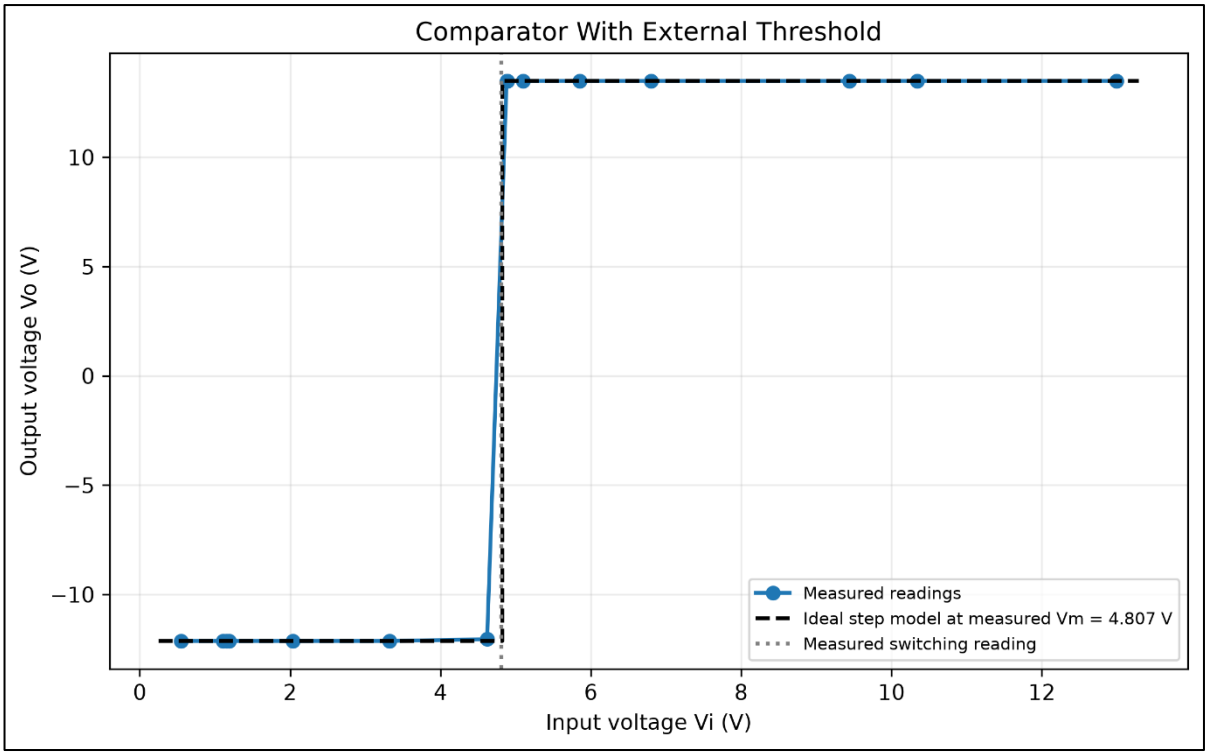
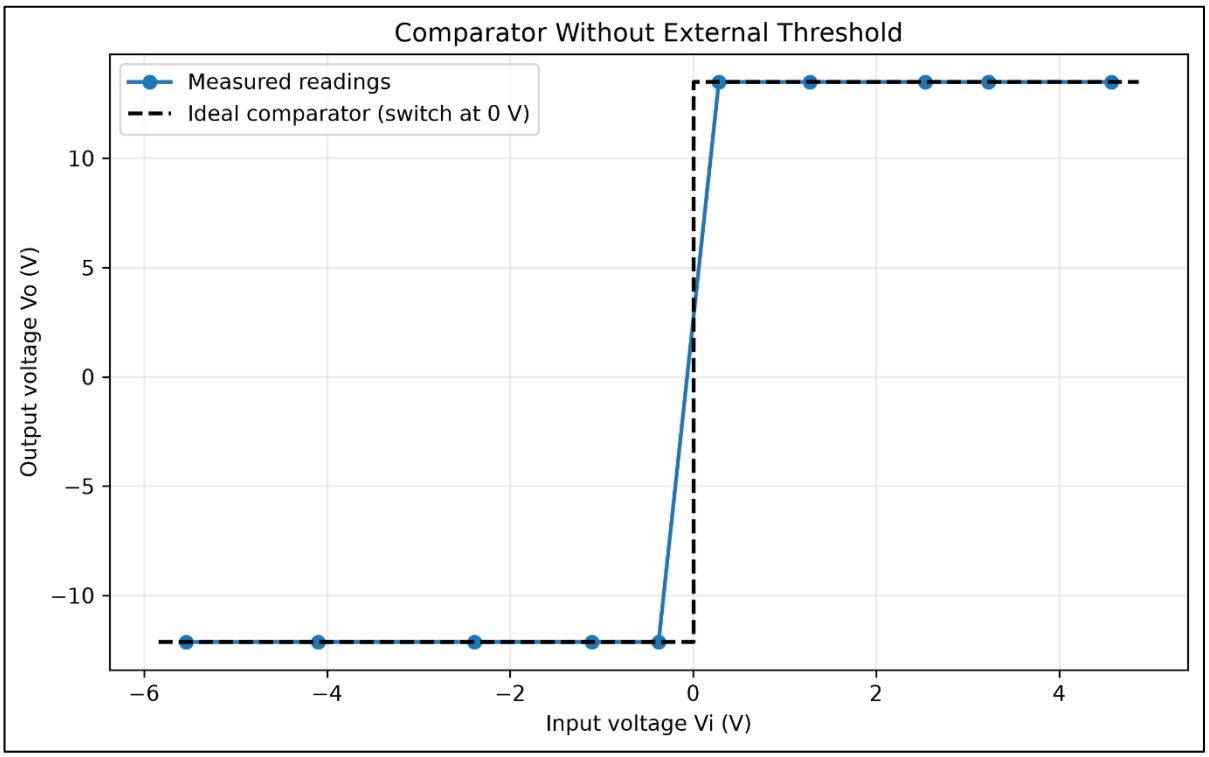
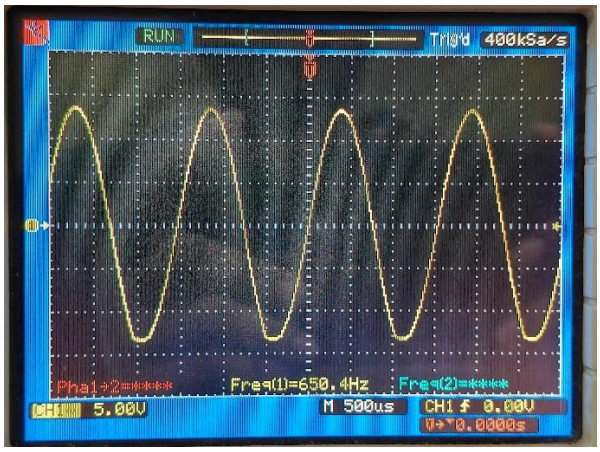
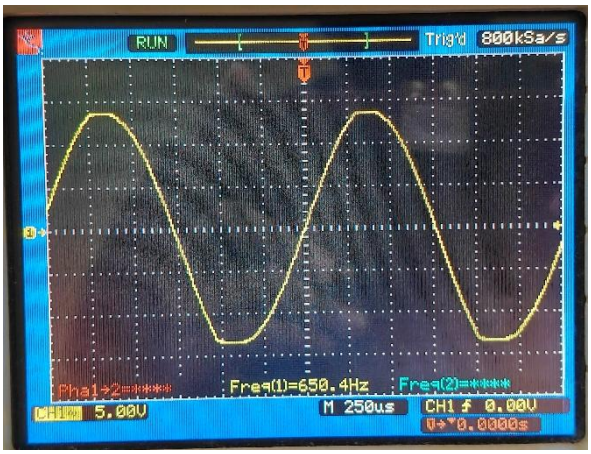
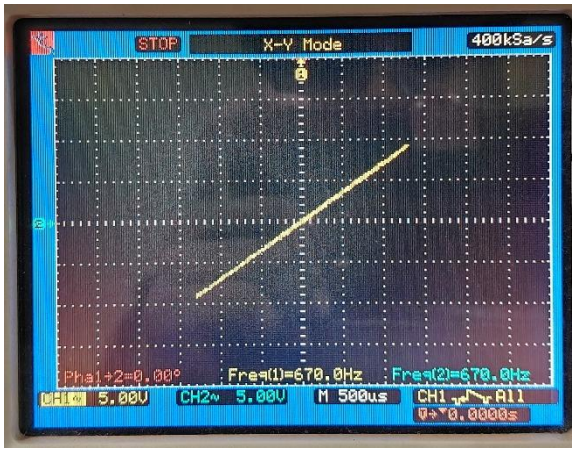
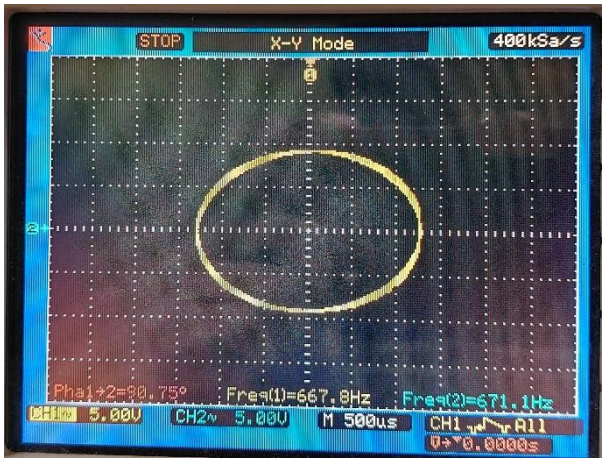

Comparator With External Threshold
Output voltage Vo (V)
Input voltage Vi (V)
Measured readings
Ideal step model at measured Vm = 4.807 V
Measured switching reading

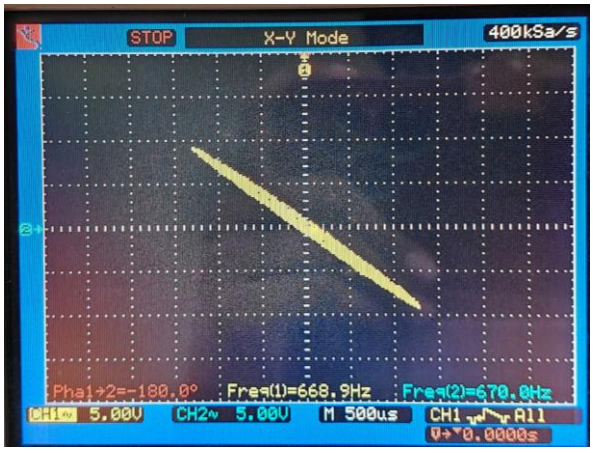

Comparator Without External Threshold
Output voltage Vo (V)
Input voltage Vi (V)
Measured readings
Ideal comparator (switch at 0 V)


RUN
Trig'd
400kSa/s
Phal+2=*****
Freq(1)=650.4Hz
Freq(2)=*****
CH1 5.00V
M 500us
CH1 ∫ 0.00V
0→▼0.0000s

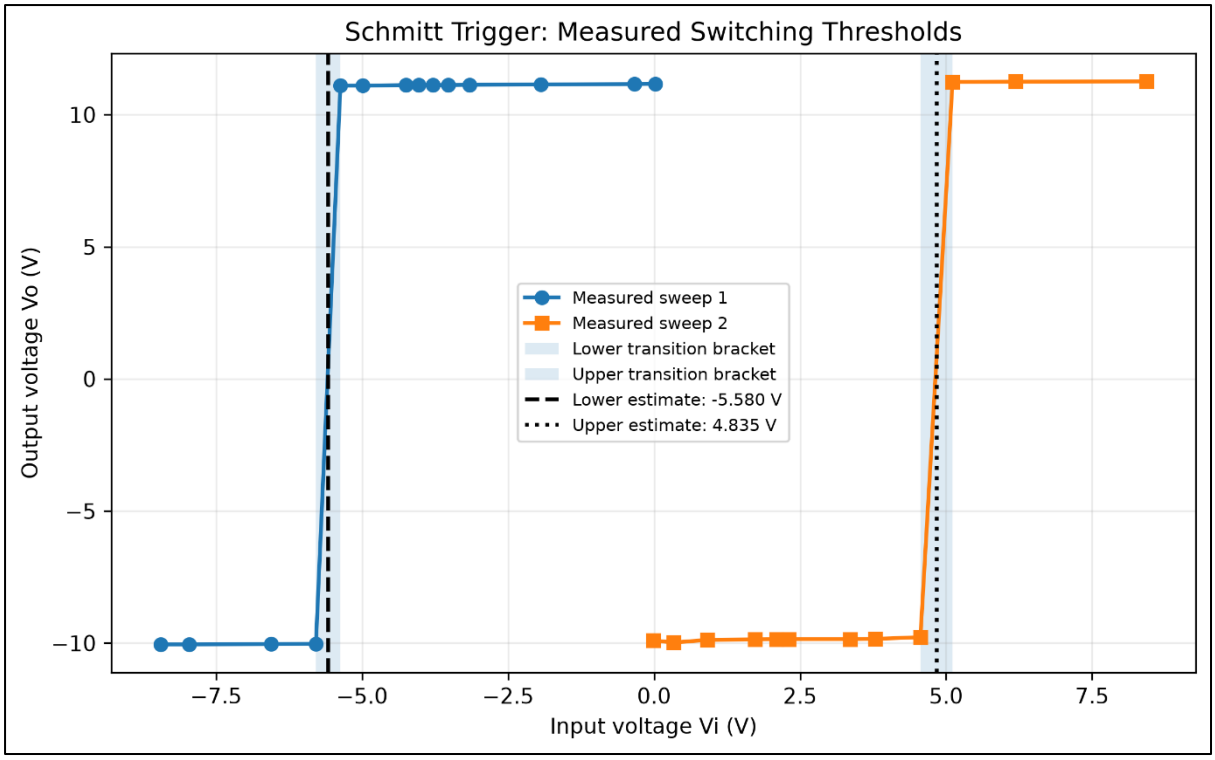

RUN
Trig'd
800kSa/s
Phal+2=*****
Freq(1)=650.4Hz
Freq(2)=*****
CH1 5.00V
M 250us
CH1 ∫ 0.00V
0→▼0.0000s

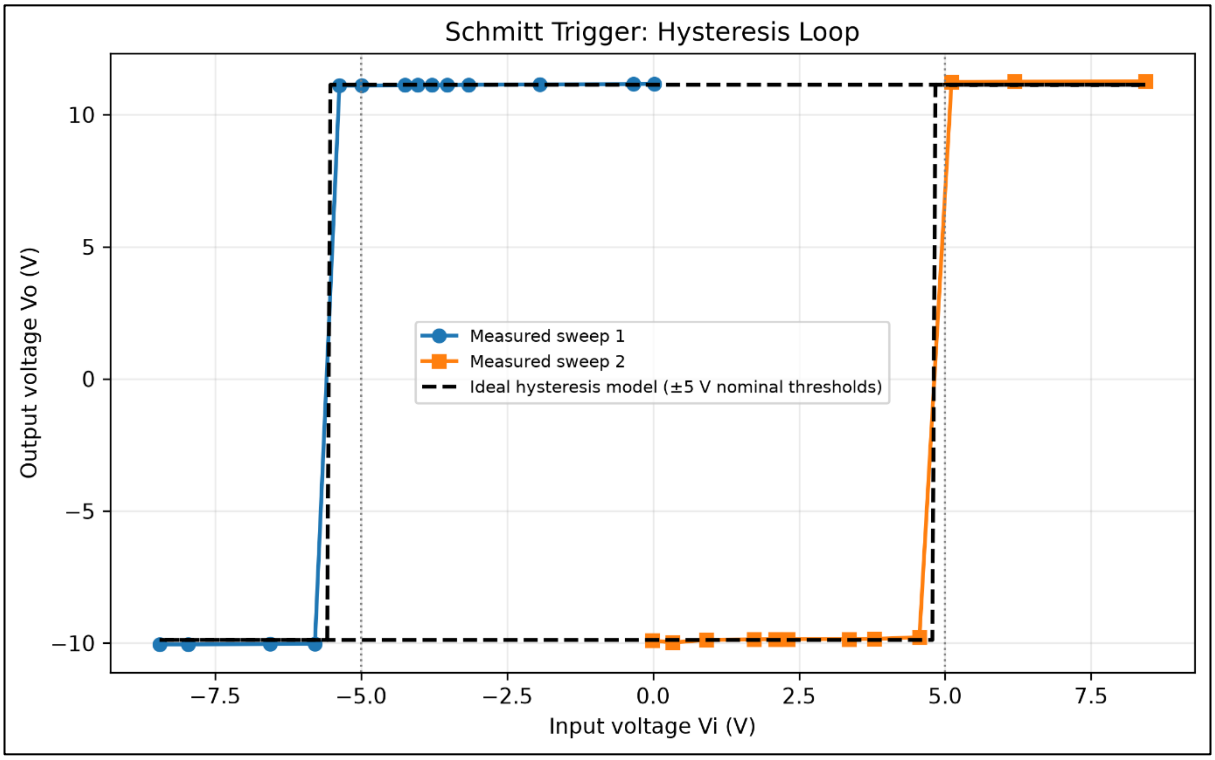

STOP
X-Y Mode
400kSa/s
Phal+2=0.00°
Freq(1)=670.0Hz
Freq(2)=670.0Hz
CH1 5.00V
CH2 5.00V
M 500us
CH1 All
0→▼0.0000s


STOP
X-Y Mode
400kSa/s
Phal+2=90.75°
Freq(1)=667.8Hz
Freq(2)=671.1Hz
CH1 5.00V
CH2 5.00V
M 500us
CH1 All
0→▼0.0000s


STOP
X-Y Mode
400kSa/s
Phal+2=-180.0°
Freq(1)=668.9Hz
Freq(2)=670.0Hz
CH1 5.00V
CH2 5.00V
M 500us
CH1 All
0→▼0.0000s


STOP
X-Y Mode
400kSa/s
Phal+2=-90.45°
Freq(1)=670.0Hz
Freq(2)=670.0Hz
CH1 5.00V
CH2 5.00V
M 500us
CH1 All
0→▼0.0000s


Schmitt Trigger: Measured Switching Thresholds
Output voltage Vo (V)
Input voltage Vi (V)
10
5
0
−5
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Measured sweep 1
Measured sweep 2
Lower transition bracket
Upper transition bracket
Lower estimate: -5.580 V
Upper estimate: 4.835 V


Schmitt Trigger: Hysteresis Loop
Output voltage Vo (V)
Input voltage Vi (V)
10
5
0
−5
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
Measured sweep 1
Measured sweep 2
Ideal hysteresis model (±5 V nominal thresholds)